FinDoc-AuditEngine v3.0 – Financial Reconciliation & Anomaly Scoring Prototype

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Business Problem in Accounts Payable (AP)

- Manual 3-way matching between Vendor Invoices, Purchase Orders (PO), and Goods Receipt Notes (GRN) is slow and error-prone.
- Static ERP rules catch exact price/quantity mismatches but miss statistical anomalies when a PO and invoice match on paper.
- Sensitive financial identifiers (PAN, Aadhaar, bank accounts, emails) risk leaking into logs or ML pipelines if unmasked.

Our Working Prototype Solution

- FastAPI reference implementation combining regex-based PII masking, deterministic 3-way matching, and unsupervised ML scoring.
- Dual-currency prototype ledger displaying both USD ($) and INR (configured at 1 USD = 86.50 INR).
- Local perturbation-based feature attribution explaining the top risk drivers behind each anomaly score.

- **Speaker Notes:** *Welcome mentors and panel members. In this first minute, I will introduce the business problem and our working prototype, FinDoc-AuditEngine v3.0.*

Why Combine Deterministic Rules with Unsupervised Anomaly Scoring?

Complementary Strengths of Rules vs. Statistical Isolation

Deterministic 3-Way Matching Rules

- Enforces configured SKU-level unit price and net accepted dock quantity (received minus rejected) rules.
- Catches explicit overbilling (e.g., billing $450/unit against a $400/unit PO contract) with exact arithmetic.
- Limitation: Cannot flag unusual payment urgency or offshore jurisdiction risk when PO and invoice numbers match.

Unsupervised Isolation Forest Indicator

- Evaluates 7 engineered features simultaneously to surface unusual transactions (e.g., $49,500 Net-1 offshore invoice).
- Trained on a reproducible seeded synthetic baseline of 1,000 transaction-like records for demonstration.
- Outputs an engineered 0–100 anomaly-risk indicator (not a calibrated probability of fraud).

- **Speaker Notes:** *Rules and anomaly detection solve different problems: deterministic matching enforces contract math, while our synthetic-baseline Isolation Forest highlights unusual multi-feature patterns.*

End-to-End 4-Stage Prototype Pipeline Architecture

Modular FastAPI Services with Stage-Level Timing Telemetry

Stages 1 & 2: Privacy & Reconciliation

- Stage 1 – FinancialPrivacyScrubber (app/core/privacy.py): Regex-based masking for 6 configured patterns (PAN, Aadhaar format, Account, Card, Phone, Email).
- Stage 2 – ReconciliationMatcher (app/services/matcher.py): Compares Invoice line items against PO rates and GRN net quantities within a $0.05 subtotal tolerance.

Stages 3 & 4: ML Scoring & Routing

- Stage 3 – AnomalyScorer (app/services/anomaly_detector.py): StandardScaler + 200-tree IsolationForest on 7 features + single-feature perturbation attribution.
- Stage 4 – ApprovalRouter & In-Memory Ledger (app/services/router.py): Assigns 1 of 5 governance tiers and appends to the prototype in-memory AUDIT_LEDGER list.

- **Speaker Notes:** *Walking through our 2-minute architecture overview: every request flows sequentially through regex PII masking, 3-way reconciliation, 7-feature Isolation Forest scoring, and governance routing.*

7-Feature Engineering & Reproducible Synthetic Baseline

How AnomalyScorer Computes the 0–100 Engineered Risk Indicator

7 Engineered Input Features

- 1. log_total_usd: ln(1 + Total_USD) to compress heavy-tailed invoice amounts.
- 2. price_variance_ratio & 3. qty_shortage_ratio: Relative rate markup and dock rejection ratios.
- 4. tax_rate_deviation (vs. 18% GST baseline) & 5. line_item_concentration.
- 6. jurisdiction_risk_weight & 7. payment_urgency_index (short Net-1/Net-2 windows).

Model Scope & Calibration Disclaimer

- Trained at startup on a seeded synthetic baseline (N=1,000, random_state=42) representing normal vs. anomalous patterns.
- Single-Feature Perturbation replaces each feature k with its synthetic baseline median to estimate marginal point impact.
- Production use would require retraining and threshold calibration on authorized historical enterprise invoice data.

- **Speaker Notes:** *Here are the 7 features fed into StandardScaler and IsolationForest. Note that this model uses a seeded synthetic baseline of 1,000 records to demonstrate anomaly scoring.*

Pydantic v2 Schema Validation, Approval Tiers & In-Memory Ledger

Fail-Fast Input Validation and 5-Tier Governance Routing

Pydantic v2 Validation & Error Handling

- Cross-field @model_validator verifies line-item sum * (1 + tax_rate) matches total_amount_usd within $0.05.
- Validates GRN rejected unitsnever exceed received units and invoice dates do not precede PO dates.
- Custom FinDocAuditException handler returns structured JSON error payloads with X-Correlation-ID.

5 Governance Tiers & Prototype Storage

- Tier 1 STP (<= $1,000, matched, low risk), Tier 2 Manager ($1k–$10k), Tier 3 Director (> $10k).
- REJECTED_DISCREPANCY (contract price/qty mismatch) & FROZEN_FRAUD_RISK (high anomaly score >= 68).
- Records are stored in an in-memory Python list (AUDIT_LEDGER) for prototype inspection and CSV export.

- **Speaker Notes:** *Before business logic runs, Pydantic v2 validates mathematical consistency. Approved or flagged decisions are stored in our prototype in-memory ledger.*

Live Demo Scenario 1: Clean Invoice (INV-1001 – Tier 1 Auto-Approval)

OfficeDepot India Pvt Ltd • $432.00 USD (INR 37,368.00) • Exact 3-Way Match

Input Payload & 3-Way Match Verification

- Vendor: OfficeDepot India Pvt Ltd (Country: IN, Terms: Net-30, Tax: 18%).
- Line Item: SKU-A101 Ergonomic Chair – 2 units billed @ $180.00 vs. PO-9001 @ $180.00 (GRN: 2 received, 0 rejected).
- Reconciliation Result: Passed ($0.00 price variance, 0 unit shortage).

ML Risk Indicator & Terminal Output

- Privacy Scrubber: Masks vendor email and PAN in pre-inference text check.
- Engineered Risk Indicator: 25.0 / 100 (LOW) – within synthetic normal baseline.
- Routing Decision: TIER_1_STP_AUTO_APPROVED (Authorized = True).

- **Speaker Notes:** *In our 5-minute live demo, Scenario 1 demonstrates a clean low-value procurement invoice (INV-1001) that passes 3-way matching and routes to Tier 1 Straight-Through Processing.*

Live Demo Scenario 2: Price Discrepancy (INV-1004 – Rejected Discrepancy)

Overcharge Technologies • $5,400.00 USD (INR 467,100.00) • Unit Rate Overbill

Contract Breach & Dynamic Table Editing

- Line Item: SKU-D404 Enterprise SSD Array – 10 units billed @ $450.00 vs. PO-9004 contract rate of $400.00/unit.
- 3-Way Matcher flags RATE_VARIANCE with an exact +$500.00 USD (INR 43,250.00) overcharge.
- In the web workbench, editing any SKU rate or GRN rejected count dynamically recomputes this variance.

Risk Attribution & Governance Action

- Engineered Risk Indicator: 52.1 / 100 (MEDIUM) driven by unit rate markup ratio (+12.5%).
- Routing Decision: REJECTED_DISCREPANCY (Authorized = False, requires HITL review).
- Demonstrates deterministic rule enforcement blocking contract overcharges immediately.

- **Speaker Notes:** *Scenario 2 shows what happens when a vendor bills $450 per unit against a $400 Purchase Order. The 3-way matcher catches the $500 overcharge and routes to REJECTED_DISCREPANCY.*

Live Demo Scenario 3: Critical-Risk Anomaly (INV-1005 – Frozen Fraud Risk)

Shadow Island Holding BZ • $49,500.00 USD (INR 4,281,750.00) • PO Matched but High Anomaly

Why Deterministic Rules Alone Pass This Invoice

- Line Item: SKU-E505 Advisory Retainer – 1 unit @ $49,500.00 matches PO-9005 ($49,500.00) with 0% tax.
- Because price and quantity match the PO, a rule-only engine would escalate to routine approval.
- However, the invoice combines an offshore jurisdiction (BZ), Net-1 rush payment demand, and near-$50k amount.

7-D Isolation Forest & Perturbation Attribution

- Engineered Risk Indicator: 99.0 / 100 (CRITICAL) – isolated in synthetic feature space.
- Top Perturbation Drivers: Jurisdiction Risk Weight, Payment Window Urgency (Net-1), and Log Capital Scale.
- Routing Decision: FROZEN_FRAUD_RISK (Authorized = False, disbursement blocked for audit).

- **Speaker Notes:** *Scenario 3 is the centerpiece of the hybrid design: INV-1005 matches its PO on price and quantity, but the Isolation Forest flags the combination of offshore jurisdiction, Net-1 urgency, and $49,500 amount.*

Automated PyTest Verification Suite (All Collected Tests Passed)

Day 29: 28 Passed (python -m pytest tests -q) • Day 30: 33 Passed (python -m pytest tests -q)

What Our Automated Tests Verify

- test_audit_engine.py (10 tests): Clean 3-way match, price variance, GRN dock shortage, anomaly monotonicity, governance tiers, regex PII masking.
- test_error_handling.py (2 tests): Custom exception hierarchy and structured JSON error responses.
- test_frontend_routes.py & test_regression.py (13 tests): Dashboard HTML/CSS/JS, invoice-text parser, batch demo, CSV export, and dynamic multi-line inputs.

Day 29 & Day 30 Route Verification

- test_mentor_prep_routes.py (3 tests): Verifies 12-slide API, 10 interview Q&As, and checklist/PDF downloads (Total = 28 passed in Day 29).
- test_day30_portfolio.py (5 tests): Verifies portfolio overview, panel tracks, 90-day roadmap, and resume PDF downloads (Total = 33 passed in Day 30).
- Note: Pass rate reflects all collected functional/API tests passing; separate line coverage can be measured via pytest-cov.

- **Speaker Notes:** *Running python -m pytest tests -q executes 28 tests in Day 29 and 33 tests in Day 30, with all collected tests passing across matching math, regex masking, and API endpoints.*

Key Engineering Challenges & How We Addressed Them

Practical Lessons from Building and Refining the Prototype

Challenge 1: Decoupling UI Edits from Static Presets

- Issue: Initial frontend prototypes bound invoice submission payloads to static preset PO objects, ignoring live table edits.
- Fix: Implemented collectLineItemsFromDOM() in dashboard.js so Invoice rates, PO rates, and GRN quantities are read directly from live DOM inputs.

Challenge 2: Feature Scaling & Floating-Point Math

- Issue: Raw USD totals ($432 to $49,500) dominated bounded variance ratios (0.0–1.0); binary floats caused penny rounding errors.
- Fix: Applied ln(1 + Total_USD) + StandardScaler before IsolationForest, and enforced 2-decimal rounding with a $0.05 subtotal tolerance.

- **Speaker Notes:** *Two practical engineering challenges stood out: decoupling the frontend DOM table from static presets so any user edit is evaluated live, and handling floating-point tax tolerances cleanly.*

Honest Assessment of Current Prototype Limitations

Distinguishing Prototype Capabilities from Production Requirements

Data, Storage & Model Limitations

- 1. In-Memory Audit Ledger: AUDIT_LEDGER is a Python list in RAM; records reset when the FastAPI process restarts.
- 2. Synthetic ML Baseline: The Isolation Forest is fit on 1,000 seeded synthetic records, not real company invoices; scores are uncalibrated risk indicators.

Security, Parsing & Operational Limitations

- 3. Regex PII Scope: Covers 6 configured regex patterns (PAN, Aadhaar format, etc.) rather than context-aware NER across unstructured layouts.
- 4. Invoice-Text Parser vs. True OCR: /api/v1/audit/parse-raw parses raw text strings via regex; it does not yet integrate an optical PDF/image OCR engine.
- 5. Authentication & Concurrency: No OAuth2/RBAC or distributed idempotency locks in the local prototype.

- **Speaker Notes:** *Being transparent about limitations is essential in engineering reviews: our ledger is currently in-memory, our Isolation Forest uses a synthetic baseline, and our text parser uses regex rather than optical OCR.*

Planned Production Improvements & 90-Day Learning Roadmap

Next Steps for Persistence, Model Calibration, Security & Scale

Planned Architecture Upgrades (Future Work)

- Durable Storage & Concurrency: Replace in-memory list with PostgreSQL (SQLAlchemy + asyncpg) and Redis idempotency keys.
- Production Model Validation: Evaluate against labeled historical AP datasets, calibrate thresholds, and monitor feature drift (EvidentlyAI PSI).
- Optical Document Extraction & Security: Integrate LayoutLMv3/OCR for PDF invoices and add OAuth2/JWT role-based access control.

90-Day Post-Internship Learning Plan

- Month 1 (Days 1–30): Hybrid BM25 + Dense RAG, LangGraph agent workflows, and automated RAGAS evaluation.
- Month 2 (Days 31–60): Async PostgreSQL, MLflow model registry, ONNX Runtime serving, and Kubernetes deployments.
- Month 3 (Days 61–90): Event-driven Kafka pipelines, AppSec hardening, and distributed system design.

- **Speaker Notes:** *To transition this prototype toward production, our roadmap prioritizes PostgreSQL persistence, calibration on real historical data, optical OCR integration, and RBAC security. Thank you–I welcome your questions.*